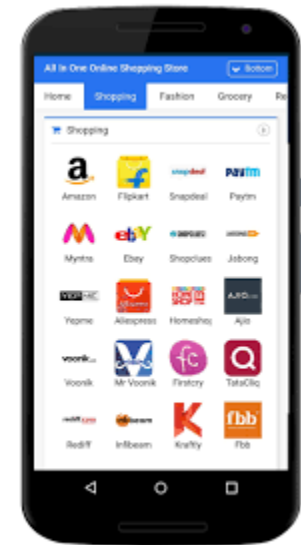




Database Concepts



Databases that you may use.....



Data

Data

- A necessity for almost any enterprise to carry out its business. Consists of raw facts, and when organized may be transformed into information

Database

- A collection of data organized to meet users' needs

Database management system (DBMS)

- A group of programs that manipulate the database and provide an interface between the database and the user of the database or other application programs



DBMS 'Discussion'

A collection of programs that enables you to store, modify, and extract information from a database. There are many different types of DBMSs, ranging from small systems that run on personal computers to huge systems that run on mainframes.

Examples:

- computerized library systems
- automated teller machines
- flight reservation systems
- computerized parts inventory systems



Hierarchy of Data

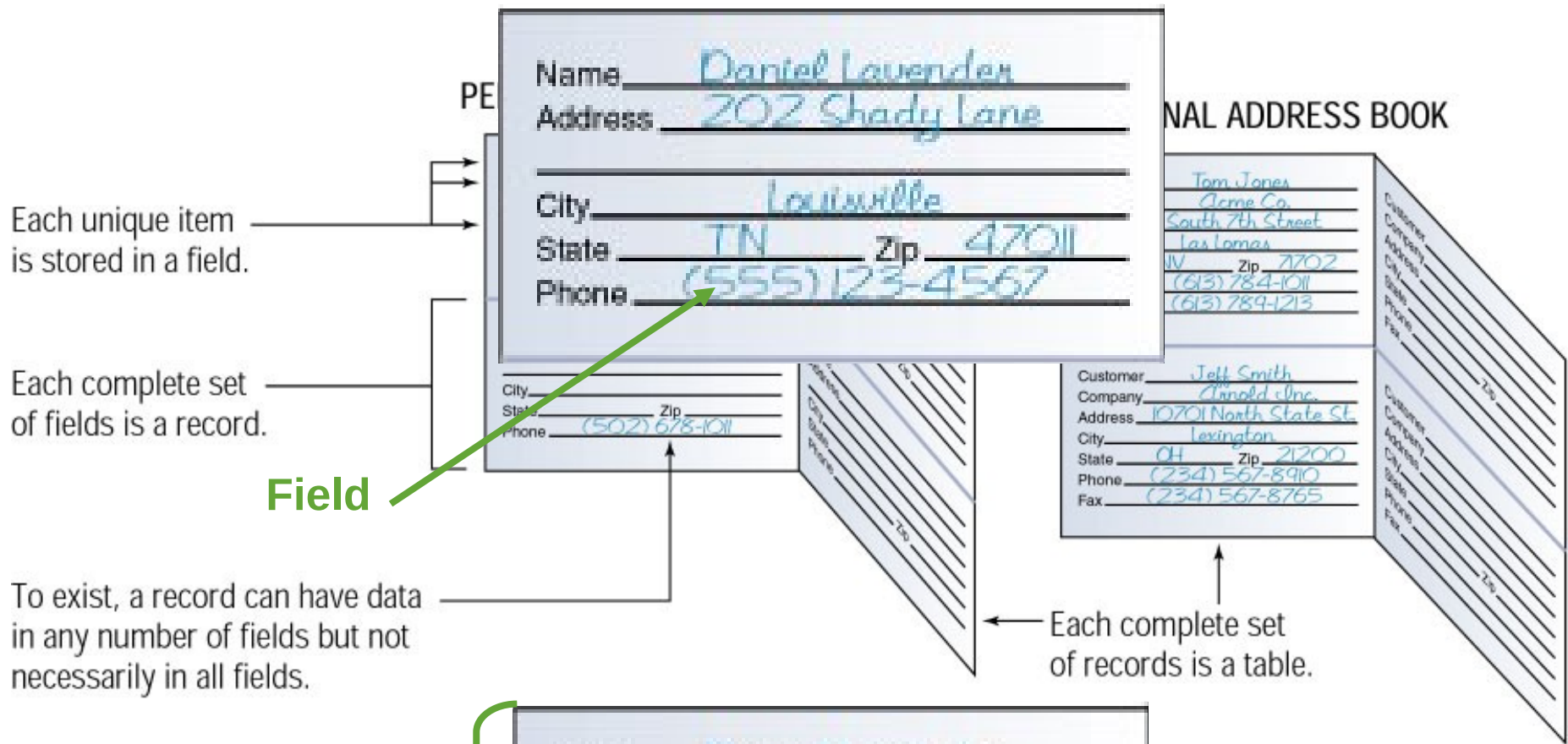


Basics of Data Arrangement and Access

The Data Hierarchy

- Recall...8 bits => 1 byte => 1 character
- **Field** - a logical grouping of characters into a word, a small group of words, or a complete number
- **Record** - a logical grouping of related fields
- **File** - a logical grouping of related records
- **Database** - a logical grouping of related files



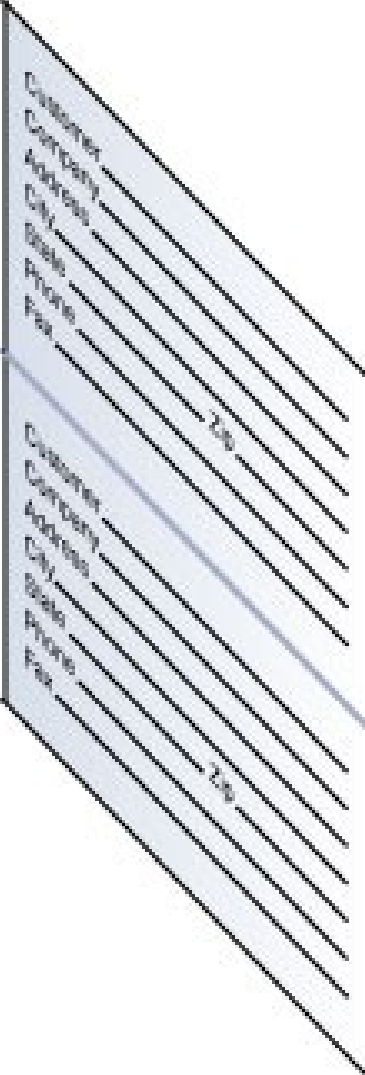


Record

PROFESSIONAL ADDRESS BOOK

Customer	Tom Jones	
Company	Acme Co.	
Address	11 South 7th Street	
City	Las Vegas	
State	NV	Zip 7702
Phone	(613) 784-1011	
Fax	(613) 784-1213	

Customer	Jeff Smith	
Company	Donald Inc.	
Address	10701 North State St	
City	Lexington	
State	OH	Zip 21200
Phone	(234) 567-8910	
Fax	(234) 567-8765	



File/Table

The Hierarchy of Data

Hierarchy of data

Example

Database

Personel file

Department file

Payroll file

(Project database)

Files

005-10-6321 Johns Francine 10-7-65
549-77-1001 Buckley Bill 2-17-79
098-40-1370 Fiske Steven 1-5-85

(Personnel file)

Records

098-40-1370 Fiske Steven 1-5-85 598

(Record containing
SSN, last name,
first name, date of
hire)

Fields

Fiske

(Last name field)

**Characters
(bytes)**

1000100

(Letter 'F' in ASCII)

Data Entities, Attributes, and Keys

Entity

- A generalized class of people, places, or things (objects) for which data are collected, stored, and maintained
- E.g., Customer, Employee

Attribute

- A characteristic of an entity; something the entity is identified by
- E.g., Customer name, Employee name

Keys

- A field or set of fields in a record that is used to identify the record
- E.g., A field or set of fields that uniquely identifies the record



Keys and Attributes

Employee #	Last name	First name	Hire date	Dept. #
005-10-6321	Johns	Francine	10-7-65	257
549-77-1001	Buckley	Bill	2-17-79	650
098-40-1370	Fiske	Steven	1-5-85	598

Key field

Attributes (fields)

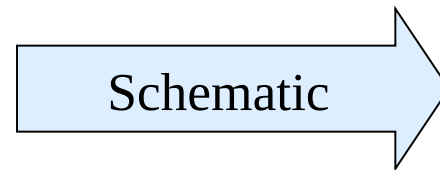
Entities
(records)



The Traditional Approach

The traditional approach...

- Separate files are created and stored for each application program



TRADITIONAL FILE SYSTEM

Same data is stored in different files (Redundancy)

Admission File

Roll No	Name	Address	Phone No
101	Rahul	Thrissur	9876543210
102	Anu	Kottayam	9123456780
103	Nikhil	Kochi	9988776655
...	...	...	...

Student File

Roll No	Name	Address	Phone No	Course Enrolled
101	Rahul	Ernakulam	9876543210	BCA
102	Anu	Kottayam	9123456780	BCA
103	Nikhil	Kochi	9988776655	BSc CS
...	...	...	...	...

Library File

Roll No	Name	Address	Book Issued
101	Rahul	Ernakulam	DBMS
102	Anu	Kottayam	OS
103	Nikhil	Kochi	CN
...	...	...	...

Examination File

Roll No	Name	Address	Marks
101	Rahul	Ernakulam	85
102	Anu	Kottayam	78
103	Nikhil	Kochi	92
...	...	...	...

Example:
Rahul changes his address
to "Thrissur"



Admission File

Address updated ✓
(Thrissur)

Student File

Address not updated ✗
(Ernakulam)

Library File

Address not updated ✗
(Ernakulam)

Examination File

Address not updated ✗
(Ernakulam)

Problems in File System

- Data Redundancy
- Data Inconsistency
- Data Isolation
- Difficulty in Access
- Lack of Security
- No Integrity
- No Concurrency Control
- No Backup & Recovery
- Program-Data Dependence

Drawbacks

- **Data redundancy**
 - The same data may be stored in multiple files, wasting storage space.
- **Data inconsistency**
 - When duplicate data is updated in one file but not in others, inconsistent information results.
- **Difficulty in accessing data**
 - Retrieving specific information often requires writing a separate program.
- **Data isolation**
 - Data is scattered across different files and formats, making it difficult to retrieve combined information
- **Integrity problems**
 - File systems do not automatically enforce rules to maintain correct data.
 - Integrity constraints (e.g. account balance > 0) become “buried” in program code rather than being stated explicitly
 - Hard to add new constraints or change existing ones



Drawbacks of using file systems (cont.)

- **Concurrent access by multiple users**
 - Multiple users accessing the same file simultaneously may overwrite each other's changes.
 - Uncontrolled concurrent accesses can lead to inconsistencies
- **Security problems**
 - File systems provide limited access control, making unauthorized access easier.
 - Hard to provide user access to some, but not all, data
- **Program–Data Dependence**
 - Changes to the file structure require changes in all application programs using that file.
- **No Backup and Recovery Mechanism**
 - Recovering lost data after a system crash is difficult.

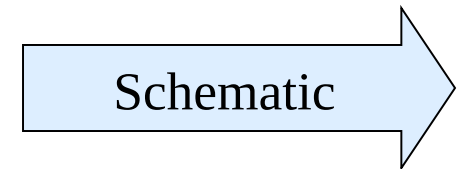
Database systems offer solutions to all the above problems



Database Approach

The database approach...

- A pool of related data is shared by multiple application programs
- Rather than having separate data files, each application uses a collection of data that is either joined or related in the database



DATABASE MANAGEMENT SYSTEM (DBMS)

Data is stored once and used by multiple applications (No Redundancy)

DATABASE

Admission Table

Student Table

Library Table

Examination Table

Admission Table

Roll No	Name	Address	Phone No
101	Rahul	Thrissur	9876543210
102	Anu	Kottayam	9123456780
103	Nikhil	Kochi	9988776655
...	...	...	...

Student Table

Roll No	Course Enrolled
101	BCA
102	BCA
103	BSc CS
...	...

Library Table

Roll No	Book Issued	Issue Date
101	DBMS	01-04-2024
102	OS	03-04-2024
103	CN	05-04-2024
...	...	...

Examination Table

Roll No	Marks
101	85
102	78
103	92
...	...

Update once in ADMISSION TABLE
(Address changed to "Thrissur")



Reflected everywhere
(Consistent Data)

Advantages of DBMS

- Reduced Redundancy
- Improved Consistency
- Data Integrity
- Security
- Concurrency Control
- Backup & Recovery
- Data Independence
- Easy Data Access

Database Management System (DBMS)

DBMS contains information about a particular enterprise

- Collection of interrelated data
- Set of programs to access the data
- An environment that is both *convenient* and *efficient* to use

Database Applications:

- Banking: all transactions
- Airlines: reservations, schedules
- Universities: registration, grades
- Sales: customers, products, purchases
- Online retailers: order tracking, customized recommendations
- Manufacturing: production, inventory, orders, supply chain
- Human resources: employee records, salaries, tax deductions

Databases touch all aspects of our lives





Why Use a DBMS?



- Data independence and efficient access.
- Reduced application development time.
- Data integrity and security.
- Uniform data administration.
- Concurrent access, recovery from crashes.



Levels of Abstraction

Physical level: describes how a record is stored.

Logical level: describes data stored in database, and the relationships among the data.

```
type customer = record
```

```
    customer_id : string;  
    customer_name : string;  
    customer_street : string;  
    customer_city : integer;
```

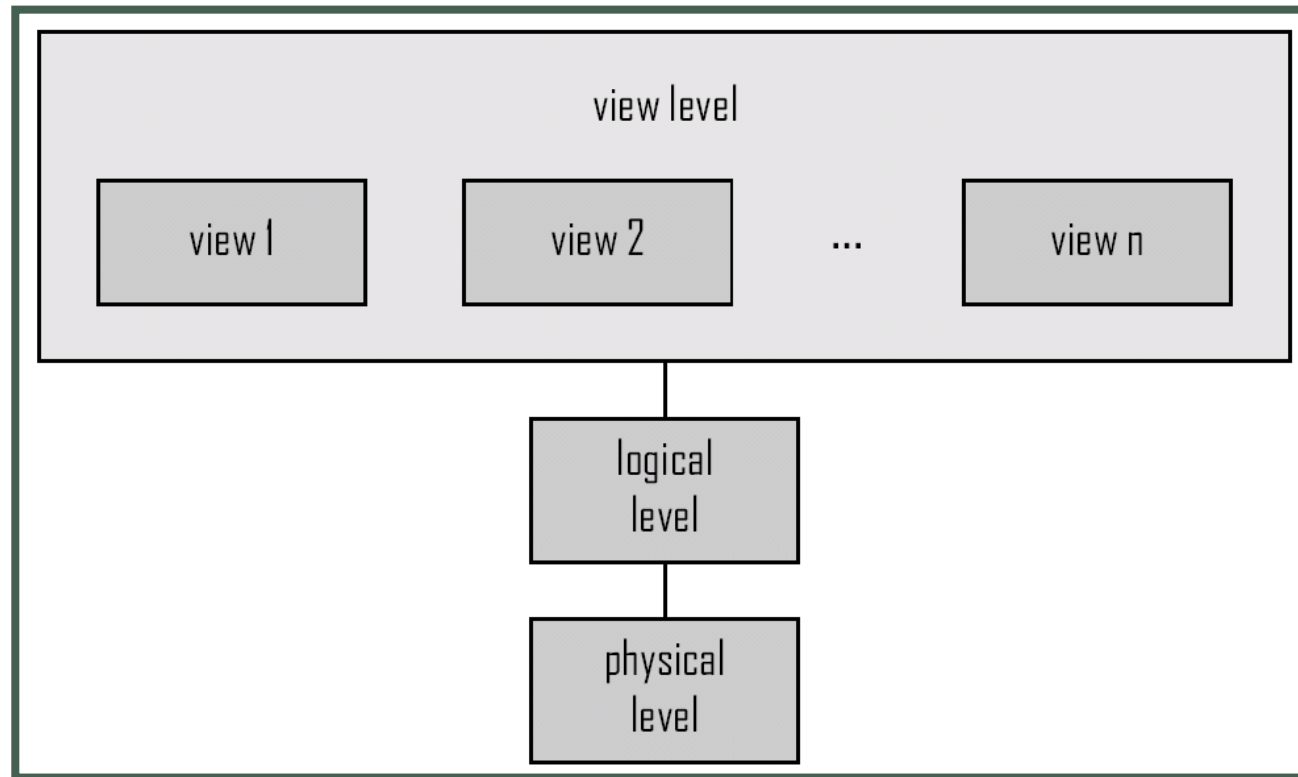
```
end;
```

View level: application programs hide details of data types. Views can also hide information (such as an employee's salary) for security purposes.



View of Data

An architecture for a database system



Instances and Schemas

Similar to types and variables in programming languages

Schema – the logical structure of the database

- Example: The database consists of information about a set of customers and accounts and the relationship between them)
- Analogous to type information of a variable in a program
- **Physical schema:** database design at the physical level
- **Logical schema:** database design at the logical level

Instance – the actual content of the database at a particular point in time

- Analogous to the value of a variable

Physical Data Independence – the ability to modify the physical schema without changing the logical schema

- Applications depend on the logical schema



Database Users

Users are differentiated by the way they expect to interact with the system

Database Administrators

Application programmers – interact with system through DML calls

Sophisticated users – form requests in a database query language

Naïve users – invoke one of the permanent application programs that have been written previously

- Examples, people accessing database over the web, bank tellers, clerical staff



Database Administrator

Coordinates all the activities of the database system; the database administrator has a good understanding of the enterprise's information resources and needs.

Database administrator's duties include:

- Schema definition
- Storage structure and access method definition
- Schema and physical organization modification
- Granting user authority to access the database
- Specifying integrity constraints
- Acting as liaison with users
- Monitoring performance and responding to changes in requirements



Data Models

A collection of tools for describing

- Data
- Data relationships
- Data semantics
- Data constraints

Relational model

Entity-Relationship data model (mainly for database design)

Other models

- *Object-based data models (Object-oriented and Object-relational)*
- *Semi-structured data model (XML)*

Other older models:-

- *Network Model*
- *Hierarchical Model*



Database Languages

- Data Definition Language
- Data Manipulation Language
- Data Control Language



Data Definition Language (DDL)

Specification notation for defining the database schema

Example: **create table** *account* (
 account-number **char**(10),
 balance **integer**)

DDL compiler generates a set of tables stored in a *data dictionary*

Data dictionary contains metadata (i.e., data about data)

- Database schema
- Data *storage and definition* language
 - Specifies the storage structure and access methods used
- Integrity constraints
 - Domain constraints
 - Referential integrity (**references** constraint in SQL)
- Authorization



Data Manipulation Language (DML)

Language for accessing and manipulating the data organized by the appropriate data model

- DML also known as query language

Two classes of languages

- **Declarative (nonprocedural)** – user specifies what data is required without specifying how to get those data, expresses the logic of a computation without describing its control flow. It attempts to minimize or eliminate side effects by describing what the program should accomplish, rather than describing how to go about accomplishing it.
- **Procedural** – user specifies what data is required and how to get those data

SQL is the most widely used query language



Relational Model

Example of tabular data in the relational model

**Attribu
tes**

<i>customer_id</i>	<i>customer_name</i>	<i>customer_street</i>	<i>customer_city</i>	<i>account_number</i>
192-83-7465	Johnson	12 Alma St.	Palo Alto	A-101
192-83-7465	Johnson	12 Alma St.	Palo Alto	A-201
677-89-9011	Hayes	3 Main St.	Harrison	A-102
182-73-6091	Turner	123 Putnam St.	Stamford	A-305
321-12-3123	Jones	100 Main St.	Harrison	A-217
336-66-9999	Lindsay	175 Park Ave.	Pittsfield	A-222
019-28-3746	Smith	72 North St.	Rye	A-201



A Sample Relational Database

<i>customer_id</i>	<i>customer_name</i>	<i>customer_street</i>	<i>customer_city</i>
192-83-7465	Johnson	12 Alma St.	Palo Alto
677-89-9011	Hayes	3 Main St.	Harrison
182-73-6091	Turner	123 Putnam Ave.	Stamford
321-12-3123	Jones	100 Main St.	Harrison
336-66-9999	Lindsay	175 Park Ave.	Pittsfield
019-28-3746	Smith	72 North St.	Rye

(a) The *customer* table

<i>account_number</i>	<i>balance</i>
A-101	500
A-215	700
A-102	400
A-305	350
A-201	900
A-217	750
A-222	700

(b) The *account* table

<i>customer_id</i>	<i>account_number</i>
192-83-7465	A-101
192-83-7465	A-201
019-28-3746	A-215
677-89-9011	A-102
182-73-6091	A-305
321-12-3123	A-217
336-66-9999	A-222
019-28-3746	A-201

(c) The *depositor* table

SQL

SQL: widely used non-procedural language

- Example: Find the name of the customer with customer-id 192-83-7465

```
select   customer.customer_name  
from     customer  
where    customer.customer_id = '192-83-7465'
```

Application programs generally access databases through one of

- Language extensions to allow embedded SQL
- Application program interface (e.g., ODBC/JDBC) which allow SQL queries to be sent to a database

